# ThermoX POWER BOARD: COPPER (60 x 60 mm square)

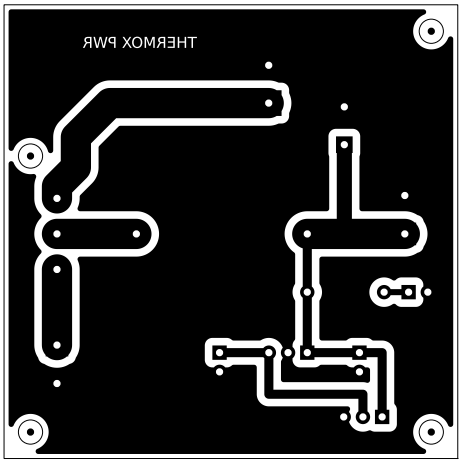
Print at 100 % / Actual size (no 'fit to page'). Check the 50 mm bar with a ruler.

Toner transfer or UV film: use this page as it is (do not mirror). Copper is on the bottom of the board.

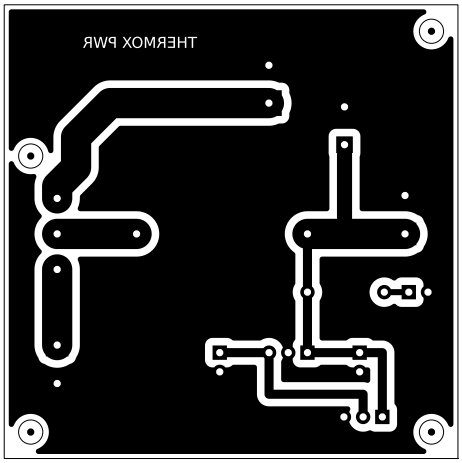
Check: the small text 'THERMOX PWR' looks mirrored on the paper and reads normally on the etched copper side.

Drill: 2.2 mm SW1, 1.5 mm F1 and M1 wire pads, 1.3 mm screw terminals, 1.0 mm C1, U1 and J7,
0.8 mm R1, LED1, C2 and C3, 3.2 mm mounting holes.

The wide tracks carry up to 7 A: cover them with a layer of solder after etching.




copy 1

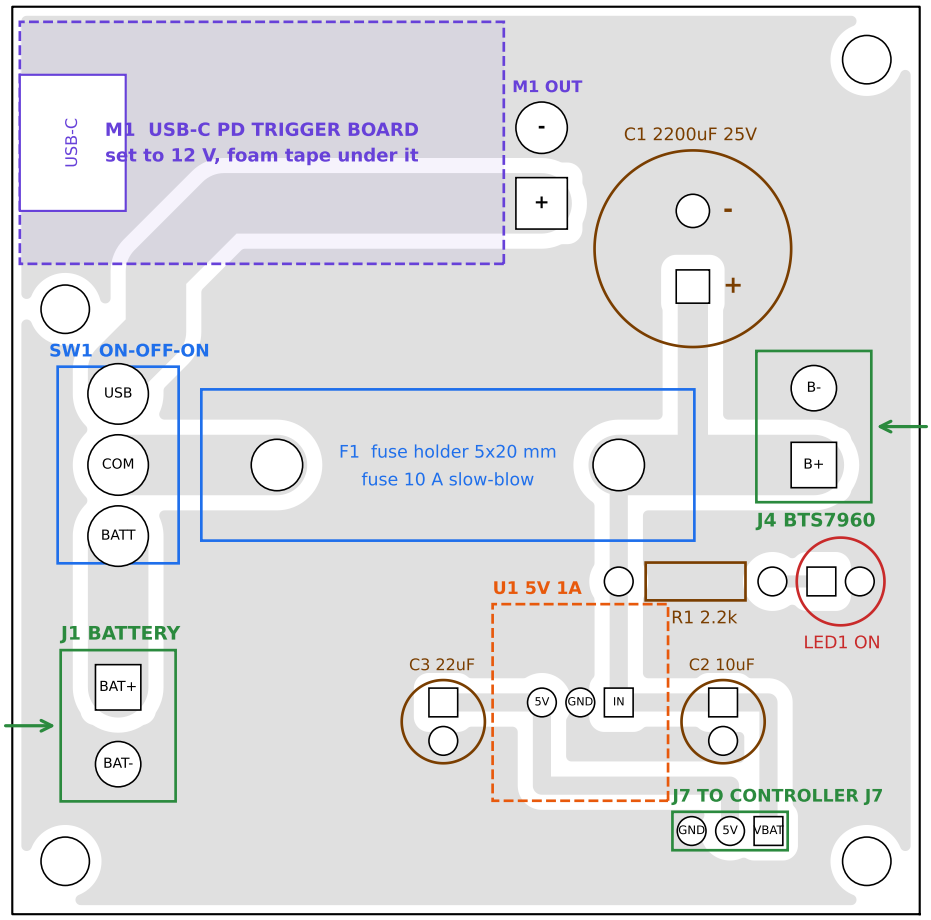



copy 2 (spare)

50 mm: measure this after printing. It must be exactly 50 mm.

# ThermoX POWER BOARD: PARTS SIDE (top view, enlarged 2 x)

Parts go on this side, copper is underneath (grey). Square pad = pin 1 (+ on C1-C3, anode on LED1, IN on U1). M1: stick the trigger board on the dashed area with thick double-sided foam tape, USB-C at the board edge. Set it to 12 V and measure it first, then solder 2 short 1 mm2 wires from its output + and - to the M1 + and - pads. SW1: the 3 lugs go through the big holes (or 3 wires to a switch on the case). U1: pin 1 (IN) in the square pad. Arrows show where the wires go into the screw terminals. C1: the stripe (-) of the capacitor towards the '-' pad.



# ThermoX POWER BOARD: WIRING, PARTS, NOTES

**CONNECTIONS**

| | |
|---|---|
| J1 BATTERY | BAT+: battery + after the BMS (P+). BAT-: battery - (BMS P-). |
| J4 BTS7960 | B+ and B- to the BTS7960 power terminals (thick wire, 1.5 mm2). M+/M- of the BTS7960 go to the Peltier. |
| J7 TO CONTROLLER | VBAT, 5V, GND to J7 VBAT, 5V, GND on the controller board (pin 1 = VBAT on both, square pad). |
| M1 (on board) | USB-C PD trigger board. Output + to the M1 + pad, output - to the M1 - pad, short 1 mm2 wires. Its USB-C end at the board edge; the charger plugs in there. |
| SW1 (on board) | ON-OFF-ON toggle: BATT = battery, middle (off), USB = trigger board. COM is the middle lug. |
| U1 (on board) | 5 V for the ESP32 and the controller board. Replaces the LM2596 module. |

**IMPORTANT**

Only ONE source at a time: SW1 picks battery or USB-C. Never connect the trigger board straight to the battery.
The trigger board does not charge the battery; on USB-C the battery is just not used.
Set the trigger board to 12 V and measure it with a meter BEFORE soldering its wires. Never 15 V or 20 V: the 12 V fan and the TEC1-12706 would be damaged. Peltier power on USB-C (phone page slider): 12V 5A charger -> 60 % max, 12V 3A -> 40 %.
SW1 (MTS-103) is rated 6 A at 125 V AC: on battery keep the Peltier power at 80 % or less, or use a 10 A toggle on 3 wires. Move the switch only when the Peltier is off (phone page: Stop) and wait in the middle position for 2 s.
Fuse F1 (10 A slow-blow, 5x20 mm) protects both sources. C1 smooths the Peltier PWM so the charger does not cut out.
U1 must be a switching type (K7805-1000R3, R-78E5.0-1.0). A plain LM7805 gets too hot here.
Before the first power-up: check with a meter that BAT+, PD+, COM, BUS and 5V are not shorted to GND. Then, with SW1 on BATT, LED1 lights, J7 VBAT reads the battery voltage and J7 5V reads 5.0 V. Only then plug in the controller.
Cover the wide copper tracks with solder after etching (they carry up to 7 A).

**PARTS**

M1: USB-C PD/QC trigger board (100W 5A type, up to 32 x 16 mm) + double-sided foam tape + 2 x 3 cm of 1 mm2 wire
SW1: ON-OFF-ON mini toggle switch, 3 lugs (MTS-103, 6 A), or a bigger 10 A ON-OFF-ON toggle on 3 wires
U1: 5 V 1 A 3-pin switching regulator: K7805-1000R3 or R-78E5.0-1.0 (same pins as a 7805)
C1: 2200 uF 25 V electrolytic, low ESR, up to 13 mm diameter, 5 mm lead spacing
C2: 10 uF 50 V (ceramic or electrolytic, 5 mm)  C3: 22 uF 16 V (ceramic or low-ESR electrolytic, 5 mm)
F1: 5x20 mm PCB fuse holder (BLX-A, pins 22.6 mm apart) + 10 A slow-blow fuse
J1, J4: 2-pin screw terminal 5.08 mm (KF301-2P)  J7: 1x3 male header 2.54 mm + 3 dupont wires to the controller J7
LED1: 3 or 5 mm LED,  R1: 2.2 kOhm 1/4 W,  M3 screws/standoffs: 4 (square) or 3 (round)

DESIGN CHECK (run by make_power_pcb.py)
clearance >= 0.6 mm between nets, ground fill 1.0 mm from other nets, all nets connected, ground path for the Peltier current >= 3.2 mm wide, no part bodies overlapping: OK

**HOW IT CONNECTS (dashed = on this board)**

